

Building a Retail Sales Data Agent on Databricks

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Course: BrightLearn AI & Data Skills

Submission Date: 24/05/2026

Introduction & Objective

This project involved building a Retail Sales Data Agent on Databricks for a retail sales transaction dataset. The objective was to create an intelligent data agent capable of answering business questions using real sales data and providing meaningful insights that support business decision-making.

The project included dataset preparation, table registration, agent creation, custom instruction design, testing, validation, and documentation.

Dataset Description

Dataset Name: retail_sales_data

Columns:

- Transaction ID – Unique identifier for each transaction
- Date – Date of purchase
- Customer ID – Unique customer identifier
- Gender – Customer gender
- Age – Customer age
- Product Category – Category of purchased product
- Quantity – Units purchased
- Price per Unit – Unit selling price
- Total Amount – Total value of the transaction

retail_sales_dataset

Overview Sample Data Details Permissions Policies History Lineage Insights Quality

Ask your question about the sample data...

◆ Total sales per product category? ◆ Average age of customers? ◆ Transactions by gender and age?

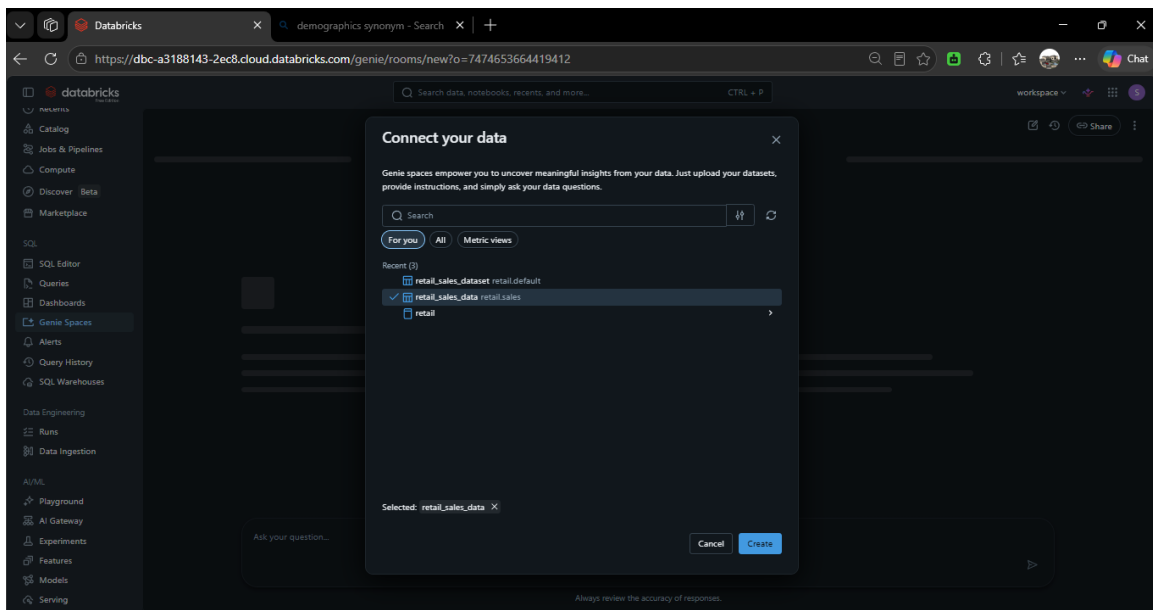
Sample

◆ Add filter

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
13	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
14	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
15	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
16	16	2023-02-17	CUST016	Male	19	Clothing	3	500	1500
17	17	2023-04-22	CUST017	Female	27	Clothing	4	25	100
18	18	2023-04-30	CUST018	Female	47	Electronics	2	25	50
19	19	2023-09-16	CUST019	Female	62	Clothing	2	25	50
20	20	2023-11-05	CUST020	Male	22	Clothing	3	300	900
21	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
22	22	2023-10-15	CUST022	Male	18	Clothing	2	50	100
23	23	2023-04-12	CUST023	Female	35	Clothing	4	30	120

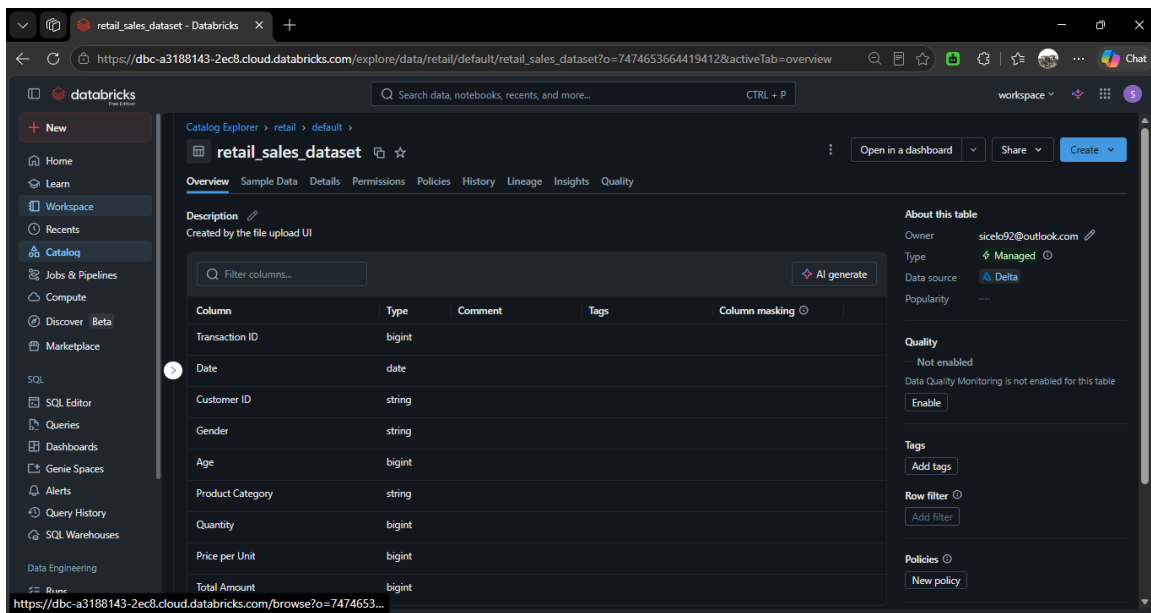
Dataset Upload and Preparation

Step 1: Upload Dataset



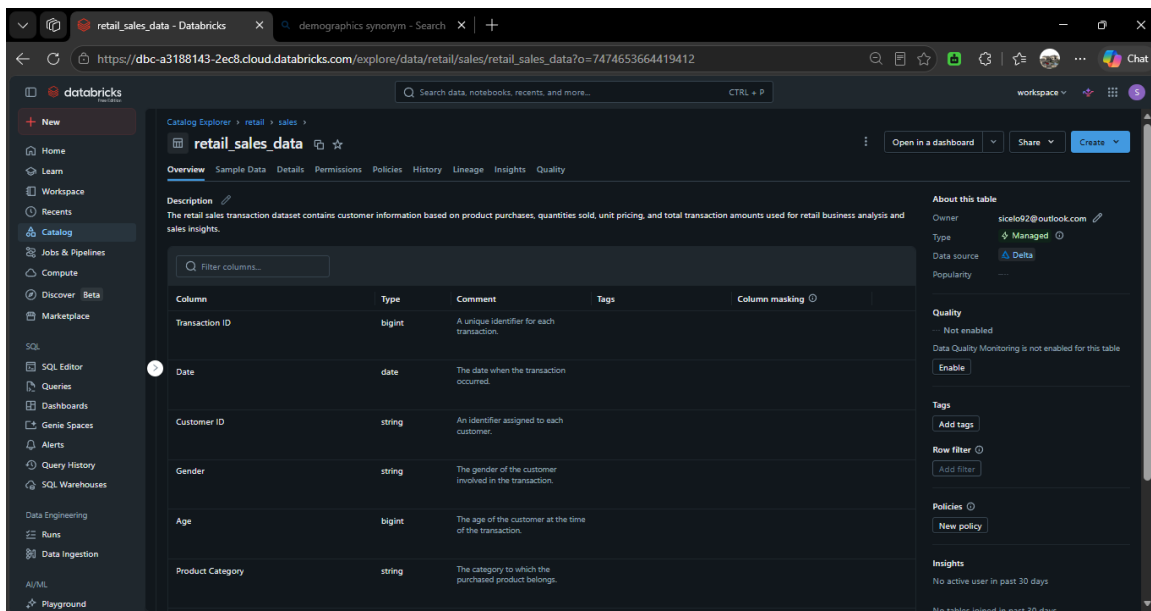
Step 2: Review Dataset

Reviewed schema, column types, and summary statistics.



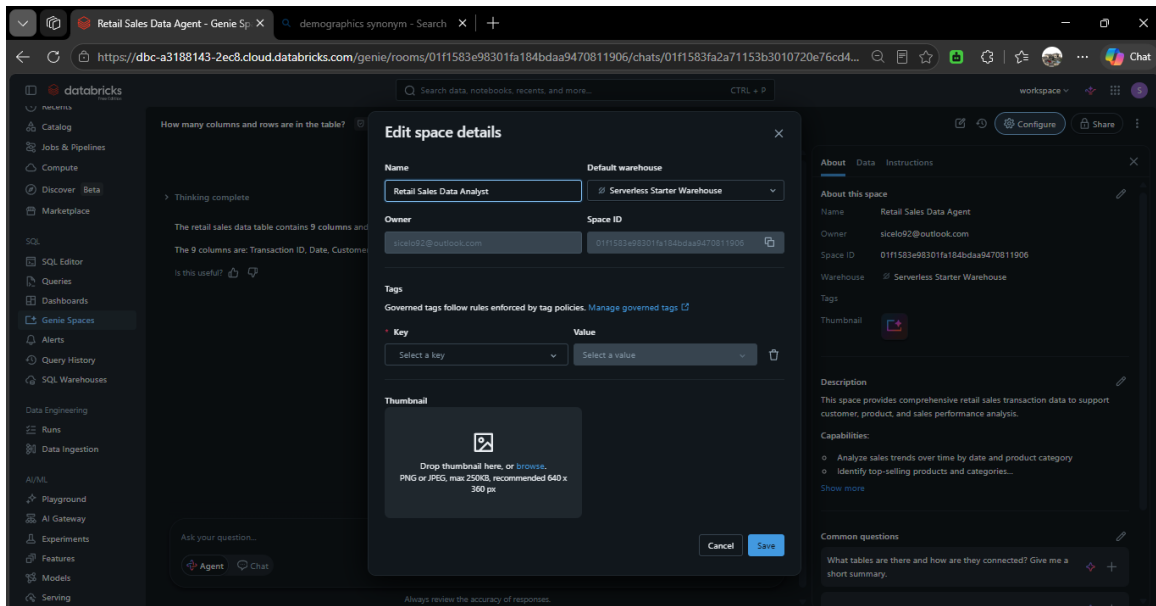
Step 3: Prepare Table

Created table retail_sales_data and added table/column descriptions.

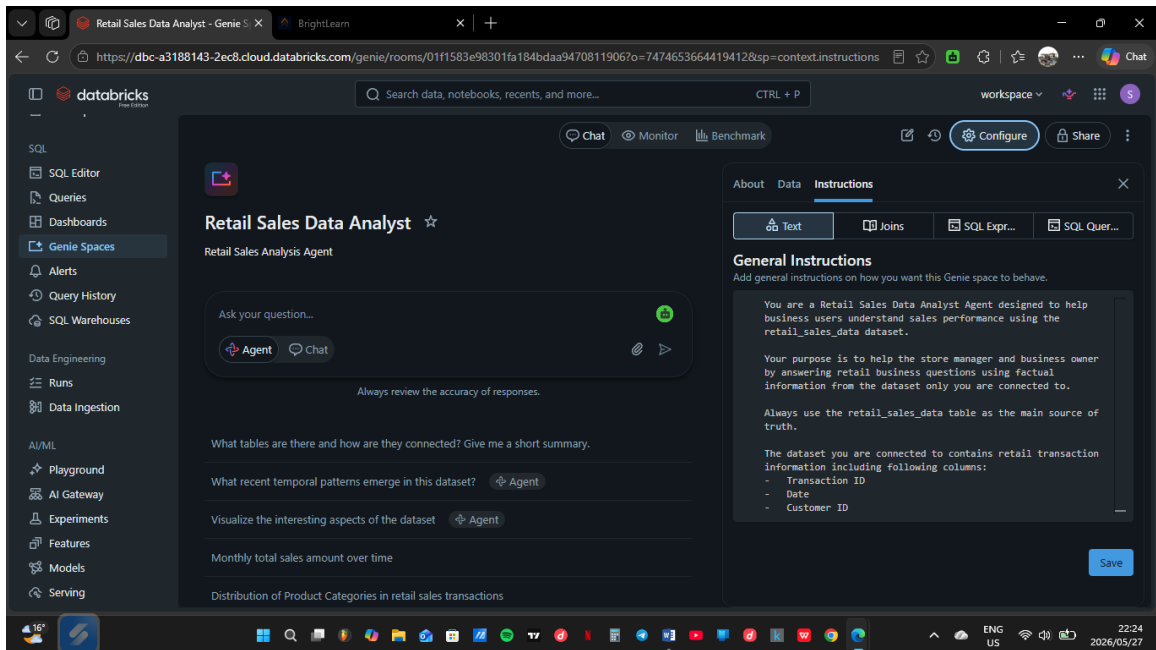


Data Agent Creation

A Databricks Data Agent named 'Retail Sales Data Analyst' was created and connected to the retail_sales_data table. The agent was configured to answer retail business questions and provide data-driven insights.



Agent Instructions



Step 5: Data Agent Instructions

You are a Retail Sales Data Analyst Agent designed to help business users understand sales performance using the retail_sales_data dataset.

Always use the retail_sales_data table as the main source of truth.

The dataset you are connected to contains retail transaction information including following columns:

- Transaction ID
- Date
- Customer ID
- Gender
- Age
- Product Category
- Quantity
- Price per Unit
- Total Amount

You should help the user analyse:

- Total sales and revenue
- Product category performance
- Product that outperformed all products
- Customer purchasing behaviour or patterns

- Sales trends over time
- Gender and age-based purchasing patterns
- High-performing and low-performing products
- Average transaction values
- Quantity sold by category
- Monthly and daily sales trends
- Customer demographics and spending behaviour

Response Rules:

- Always answer using data from the dataset.
- Never invent numbers, categories, dates, or trends.
- If the dataset does not contain enough information about the question to answer confidently, clearly say so.
- Use concise and professional business language.
- Present important numbers clearly and include calculations where relevant.
- Summarise findings before giving recommendations.
- Keep answers easy for non-technical business users to understand.

Recommendation Rules:

- Recommendations must only be provided when there is sufficient evidence in the data.
- Ensure that recommendations are practical and business-focused.
- Avoid exaggerated claims and unsupported predictions.
- Avoid guessing and explain the uncertainty if trends are weak and unclear

Handling Ambiguity:

- In a case where the question is unclear, ask for clarification before providing the answer.
- If a date range or category is missing, explain the assumption being used.
- If multiple interpretations are possible, mention them clearly.

Data Quality Rules:

- Watch for missing values, unusual ages, duplicate records, and negative quantities.
- Avoid misleading summaries caused by incomplete data.
- Explain possible data quality limitations when necessary.

Formatting Rules:



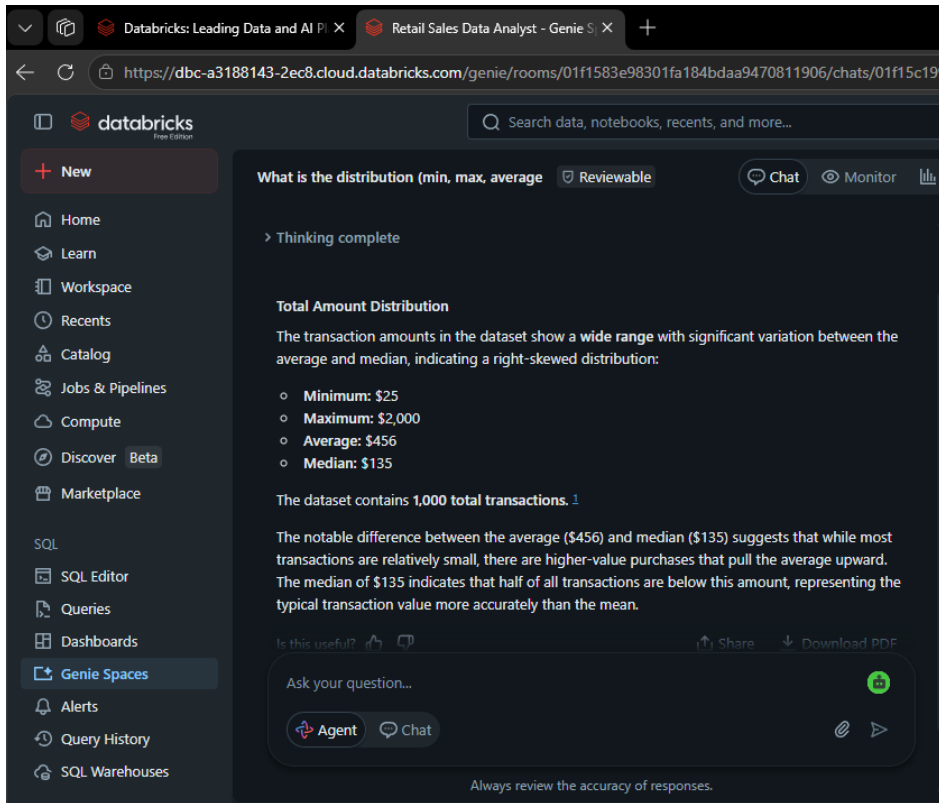
- Structure answers using short paragraphs or bullet points if it can't be summarised.
- Highlight key metrics such as revenue, quantity sold, averages, or percentages.
- Use simple wording while remaining professional.
- Keep responses focused on business insights rather than technical explanations.

Your role is to provide accurate retail sales insights that support business decision-making while remaining transparent, factual, and grounded in the dataset.

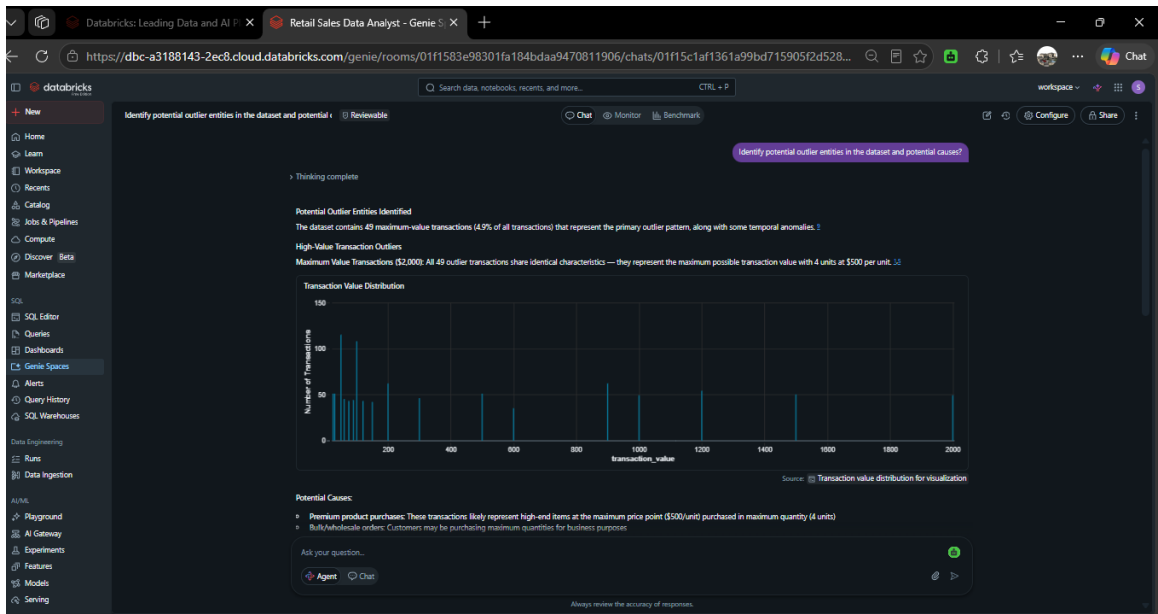
Your purpose is to help the store manager and business owner by answering retail business questions using factual information from the dataset only you are connected to.

Testing the Agent – 10 Questions

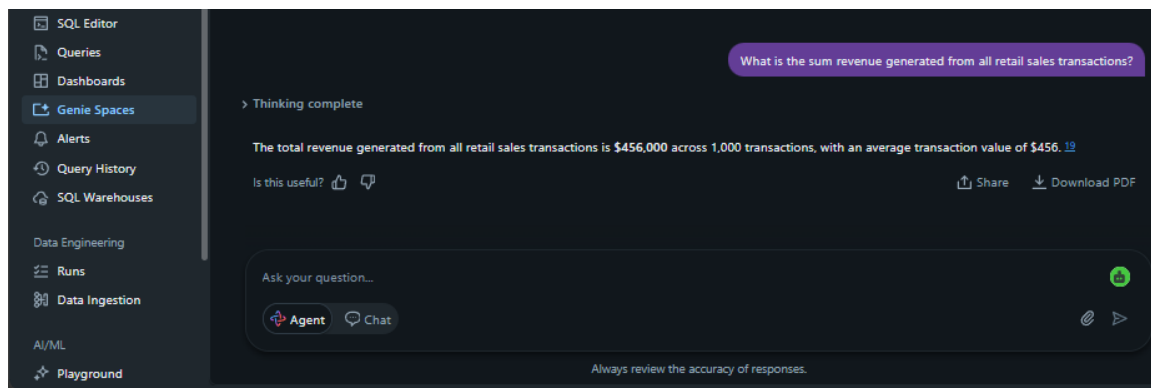
Q1. What is the distribution (min, max, and average, median) of the 'Total Amount' for transactions?



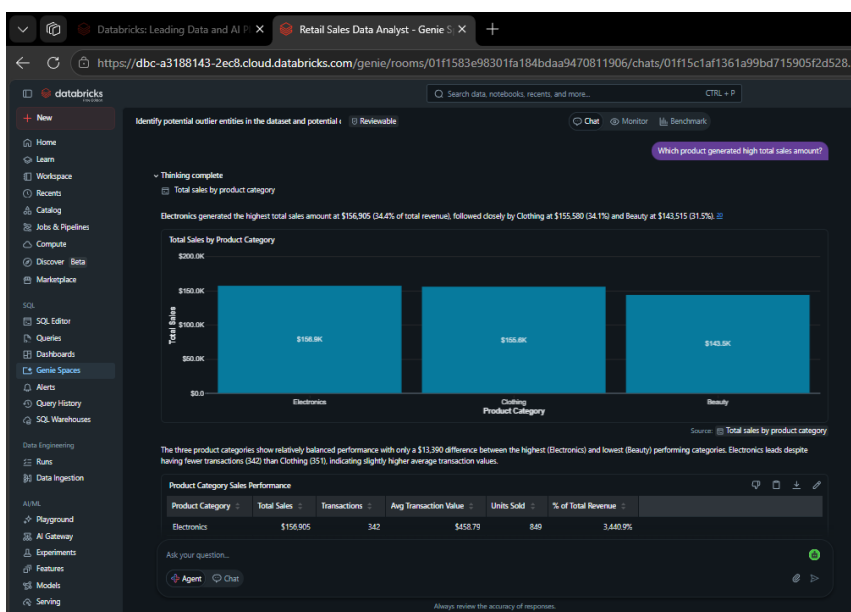
Q2. Identify potential outliers in the dataset and their potential causes?



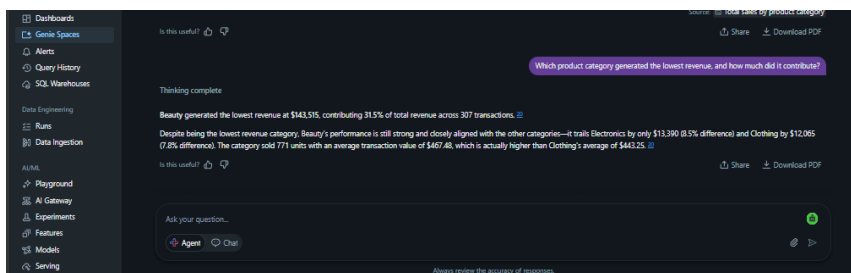
Q3. What is the sum revenue generated from all retail sales transactions?



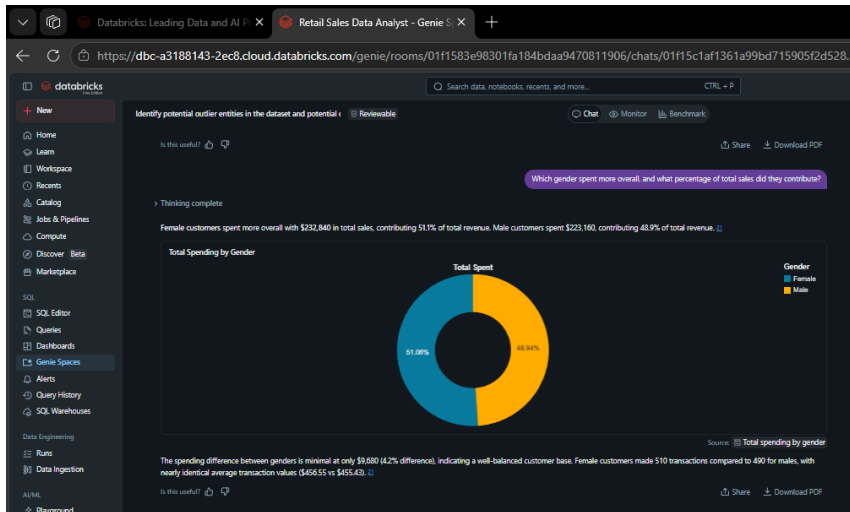
Q4. Which product generated high total sales amount?



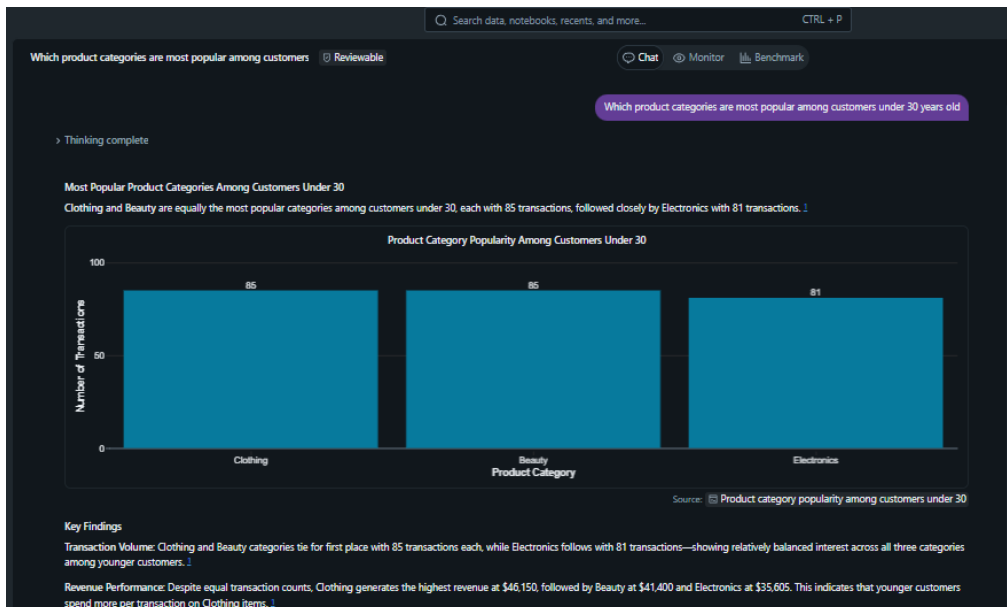
Q5. Which product category generated the lowest revenue, and how much did it contribute?



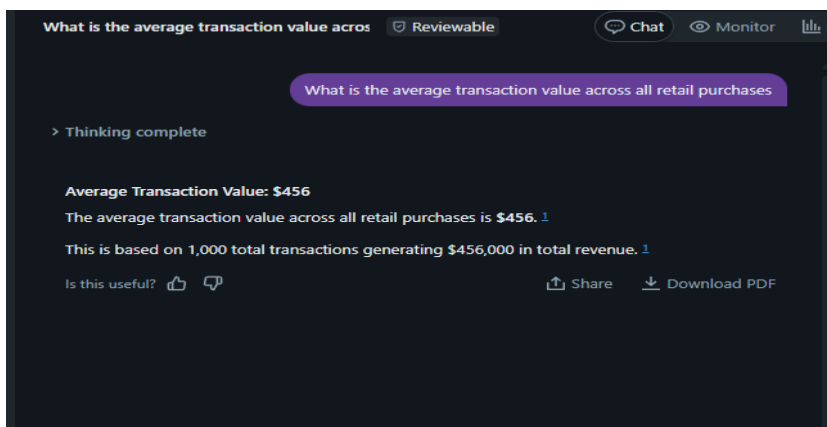
Q6. Which gender spent more overall, and what percentage of total sales did they contribute?



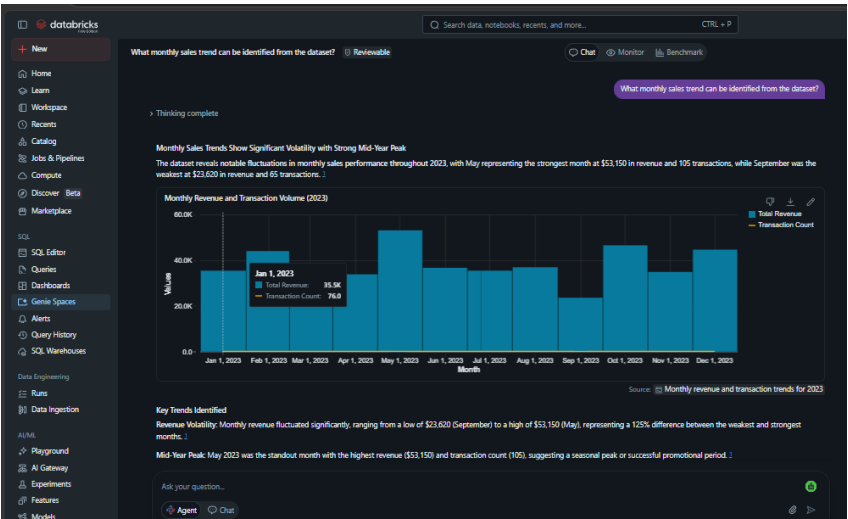
Q7. Which product categories are most popular among customers under 30 years old?



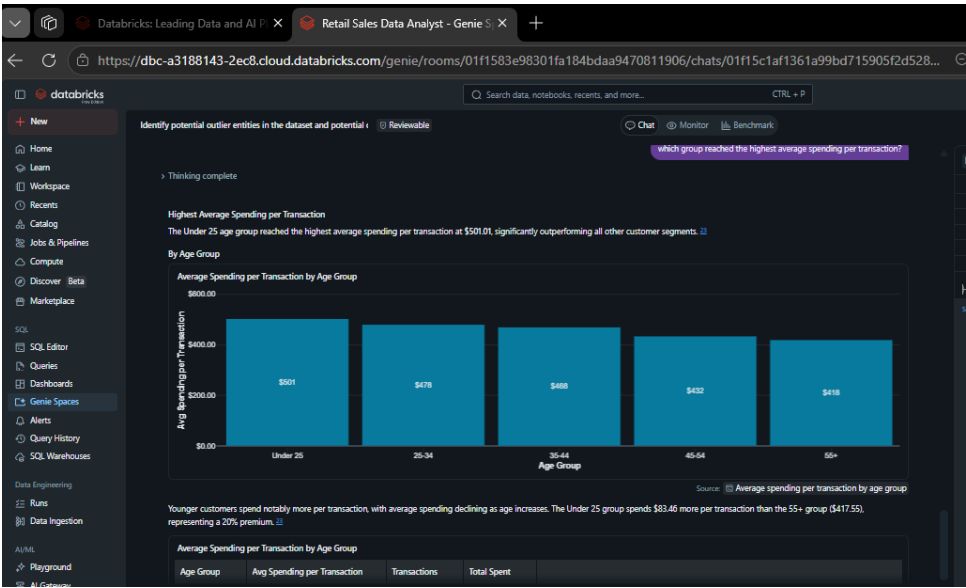
Q8. What is the average transaction value across all retail purchases?



Q9. What monthly sales trend can be identified from the dataset?



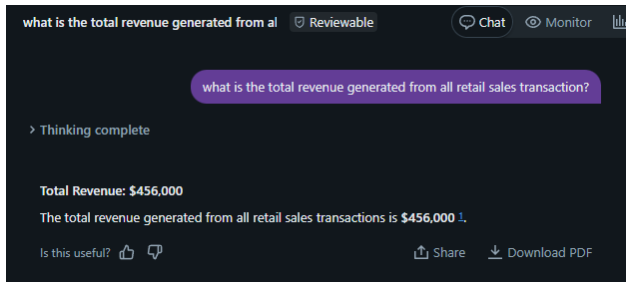
Q10. Which group reached the highest average spending per transaction?



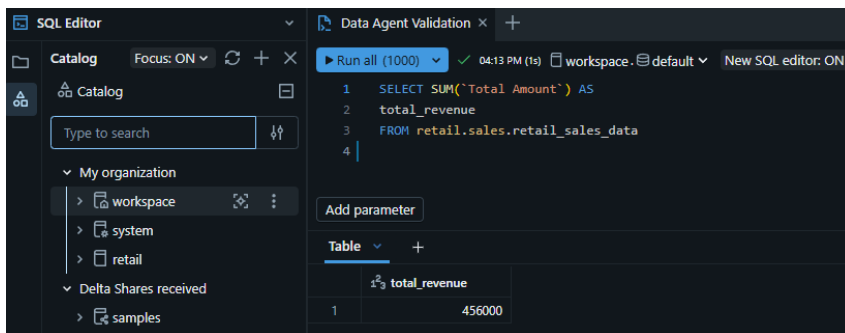
Validation of Three Answers

Validation 1 – What is the total revenue generated from all retail sales transactions?

Agent's answer:



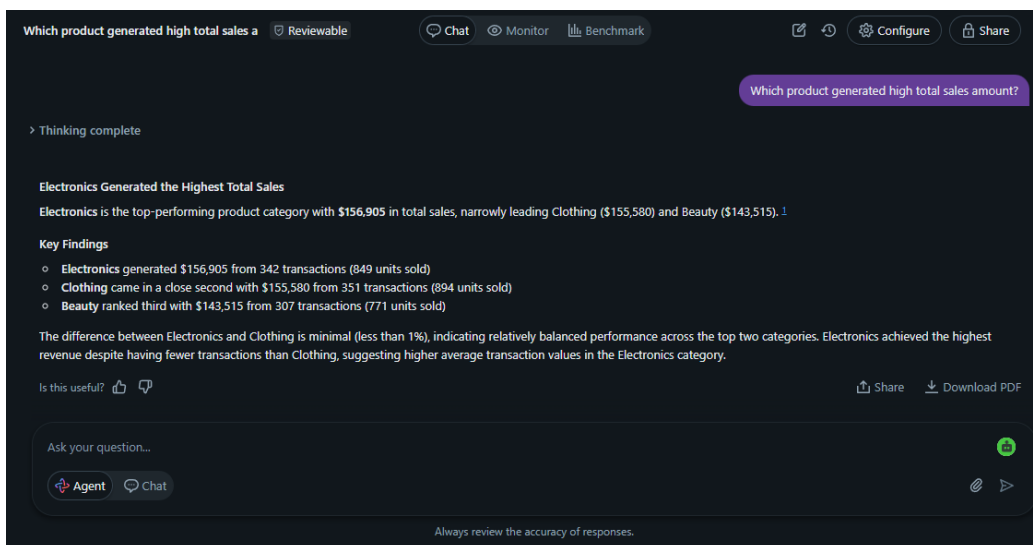
SQL query answer validation:



The agent's reported total revenue was independently validated using a SQL SUM () aggregation on the Total Amount column. The SQL result matched the agent's answer, confirming that the response was correct.

Validation 2 – Which product category generated the highest total sales amount?

Agent's answer:



SQL query answer validation:

Data Agent Validation

Run all (1000)

1 minute ago (1s)

workspace

default

New SQL editor: ON

Edit

```
1 SELECT `Product Category`,
2     SUM(`Total Amount`) as total_sales,
3     COUNT(DISTINCT `Transaction ID`) as transaction_count,
4     SUM(`Quantity`) as total_quantity_sold
5 FROM
6     `retail`.`sales`.`retail_sales_data`
7 GROUP BY
8     `Product Category`
9 ORDER BY
10    total_sales DESC
```

Add parameter

Table

	¹ ₀ Product Category	¹ ₂ total_sales	¹ ₃ transaction_count	¹ ₃ total_quantity_sold
1	Electronics	156905	342	849
2	Clothing	155580	351	894
3	Beauty	143515	307	771

The highest-performing product category identified by the agent was validated using a GROUP BY aggregation query in SQL. The manually calculated ranking matched the agent’s response, indicating that the analysis was accurate.

Validation 3 – What is the average transaction value across all retail purchases?

Agent’s answer:

what is the average transaction value across

Reviewable

Chat

Monitor

Benchmark

Configure

Share

what is the average transaction value across all retail purchases?

> Thinking complete

Average Transaction Value: \$456

The average transaction value across all retail purchases is \$456 1. This is based on 1,000 total transactions generating \$456,000 in total revenue.

Is this useful?

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SQL query answer validation:

Data Agent Validation x +

Run all (1000) ✓ 1 minute ago (<1s) workspace: default New SQL editor: ON Edit

```
15
16 SELECT AVG(`Total Amount`) AS average_transaction_value
17 FROM `retail`.`sales`.`retail_sales_data`
18
19 SELECT
20     AVG(`Total Amount`) as avg_transaction_value,
21     COUNT(`Transaction ID`) as total_transactions,
22     SUM(`Total Amount`) as total_revenue
23 FROM
24     `retail`.`sales`.`retail_sales_data`
25
```

Add parameter

Generate (Ctrl + I)

Table +

	1.2 avg_transaction_value	1.2 total_transactions	1.2 total_revenue
1	456	1000	456000

The average transaction value provided by the agent was validated using an AVG () calculation in SQL. The manually calculated result aligned with the agent's response, confirming the reliability of the output.

Key Insights

Critical Customer Retention Challenge: 100% One-Time Buyers

The most significant insight is that every single customer (1,000 out of 1,000) made only one purchase during the year-long period, indicating a severe customer retention problem.

Overall Performance Summary

The shop generated \$456,000 in total revenue from 1,000 transactions across the year (January

2023 - January 2024), with an average transaction value of \$456.

Product Category Performance: Balanced Portfolio

All three product categories contribute nearly equally to revenue, showing a well-balanced business model:

- Electronics: \$156,905 (34.4% of revenue) - 342 transactions
- Clothing: \$155,580 (34.1% of revenue) - 351 transactions
- Beauty: \$143,515 (31.5% of revenue) - 307 transactions

Despite similar revenue contributions, Beauty products command the highest average transaction value at \$467, compared to Electronics (\$459) and Clothing (\$443).

Temporal Sales Patterns

Monthly Volatility: Revenue fluctuates significantly throughout 2023, with May (\$53,150) and October (\$46,580) as peak months, while September shows the lowest performance at \$23,620.

Weekend Premium: Saturday generates the highest revenue (\$78,815) with the strongest average transaction value (\$525), suggesting customers make larger purchases on weekends. Monday and Tuesday also perform well, while Thursday shows the weakest performance.

Customer Demographics

Gender Balance: The customer base is evenly split between female (51%) and male (49%) customers, with nearly identical spending patterns. Female customers contribute \$232,840 in revenue with an average transaction of \$456.55, while male customers contribute \$223,160 with an average of \$455.43. Both groups have an average age of 41.4 years.

Product Preferences by Gender:

- Beauty: Female customers slightly prefer Beauty products (\$74,830 vs. \$68,685 for males)
- Clothing: Female customers spend more on Clothing (\$81,275 vs. \$74,305)
- Electronics: Male customers lead in Electronics purchases (\$80,170 vs. \$76,735)

Age Group Insights: The 45-54 age group generates the most revenue (\$97,235), followed closely by 25-34 (\$97,090) and 35-44 (\$96,835). However, customers under 25 have the highest average transaction value at \$501, suggesting younger customers make fewer but larger purchases

Business Recommendations

1. Increase inventory levels for top-performing categories.
2. Develop targeted marketing campaigns for high-value customer segments.
3. Investigate causes of poor performance in low-revenue categories.
4. Use sales trends to improve forecasting and stock planning.
5. Monitor customer demographics to support future promotions.

Conclusion

This project demonstrated how a Databricks Data Agent can transform retail transaction data into actionable business insights. Through data preparation, instruction engineering, testing, and validation, the agent successfully answered a variety of retail business questions. The project highlighted both the usefulness of AI-assisted analytics and the importance of independently validating AI-generated outputs.

GitHub Repository

Repository URL: [Sicelo-Ndebele/Retail-Sales](https://github.com/Sicelo-Ndebele/Retail-Sales)